

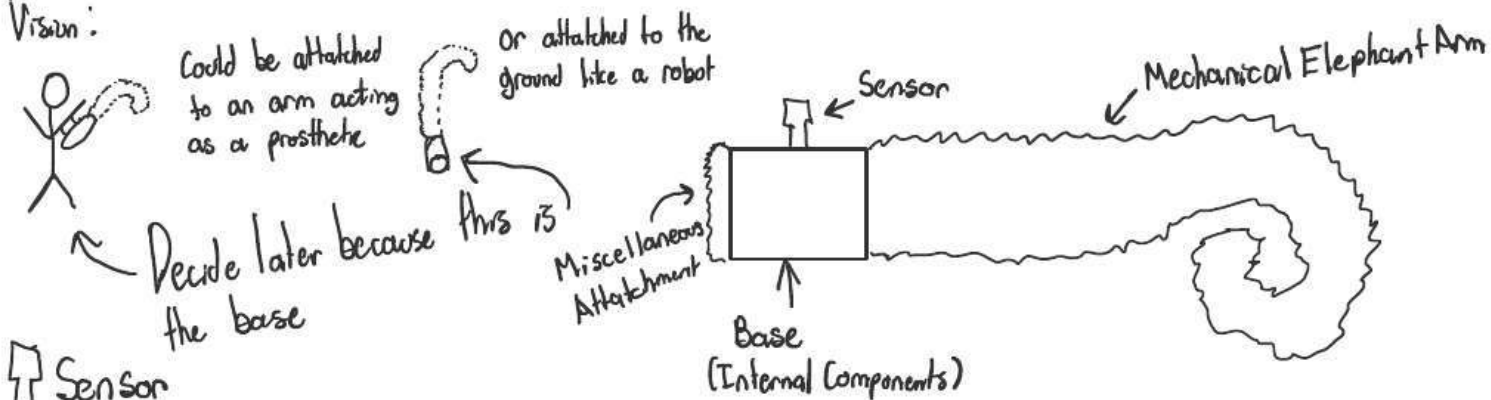
Entry 1 - ELEPHANT TRUNK REACHER - Aaron Udho

Wednesday, September 17, 2025 8:19 AM

Objective: Create an elephant trunk-like mechanical arm which is used to grab onto any kind of object, regular or irregular.

Reason: Elephants use their trunks for many daily tasks (Drinking water, picking up heavy objects, & more). This can also be useful for humans, particularly elderly whom lack muscle mass to perform these daily tasks.

Vision:



Plan: Sensor Detects the desired object and determines based on the size whether the mechanical elephant arm uses its full trunk or tip only to pick said object.

T attached inside the base or on top the base.

→ 360 camera maybe? (📷)

or this connects to a camera software with a crosshair much like in video games

where you aim towards the object & pick it up.

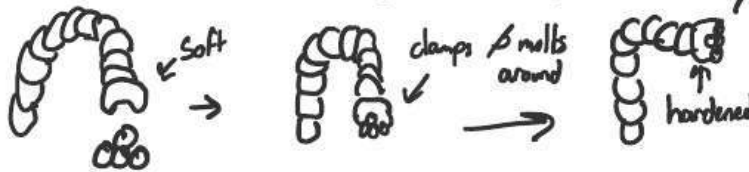
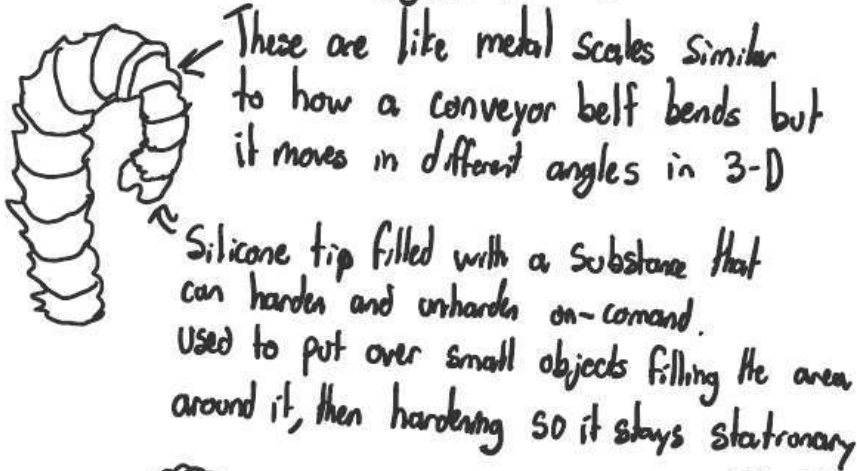
Definitely has a tracker to gauge size of desired object

Base

- The electrical parts of the machine and the power.
- provide stabilization as to not break the trunk when picking up decently heavy objects

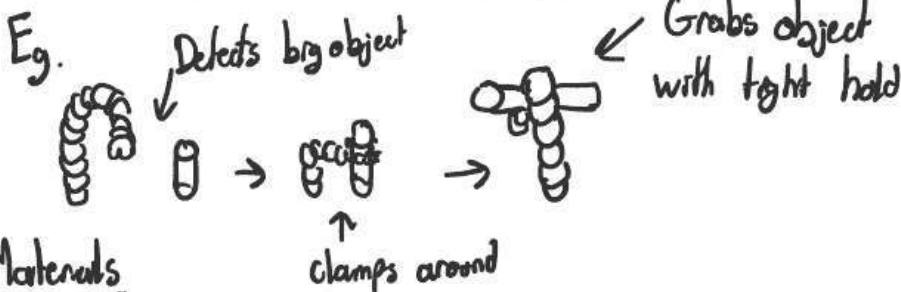
Mechanical Elephant Trunk

Plan: Sensor Sends a Signal to this and based on the Size, it does different options



For big objects

The trunk should wrap around the object in places where the object is most stable 360.



Materials

- Metal scales for aesthetics on the exterior of the trunk
- Tip is a ball that has Silicon on the outside and a substance in which hardness is possible to control inside (it)
- Inside the trunk would be fortified to prevent breakage and allowing for 3-D movement in all angles in all planes.

The Tip of the Elephant Trunk

- Contains Non-newtonian fluid
- Properties of non-newtonian fluids are is it hardens under pressure & softens when stationary
- Controls the hardness using hydraulics

→ ⊗ ← ⊗
Hardens Softens

We need to figure out a way to make tip almost liquid on pickup for small objects

This is for the following reasons:

- less damage for fragile objects
- as the liquid solidifies, it allows for more grip and stability as liquid takes up the volume of the container it's in

In order to do this we need to find the force required to turn a non-newtonian fluid more fluid

Hypothesis: In Fluid mechanics, μ (viscosity) = μ_0 (viscosity @ 0°C) · $\left(\frac{1}{1 + \alpha T + \beta T^2} \right)$

∴ $\mu \propto \sqrt{T}$, which makes it an inversely proportional relationship between viscosity and temperature.

Increasing the temperature decreases the viscosity, which increases flow rate, so by increasing or decreasing temperature, we can control the hardness of the non-newtonian fluid.

↗

wouldn't this principle apply to all fluids though?

Let's use this idea for our silicone tip

Continuation of Elephant Trunk tipIdeas for the design to control fluid

1. Force control with hydraulics

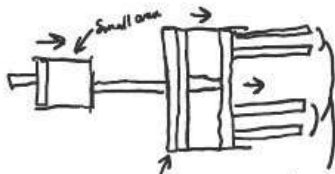
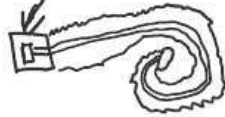
Pascals law of fluids

$$P_1 = \frac{F_1}{A_1} \quad P_2 = \frac{F_2}{A_2}$$

$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

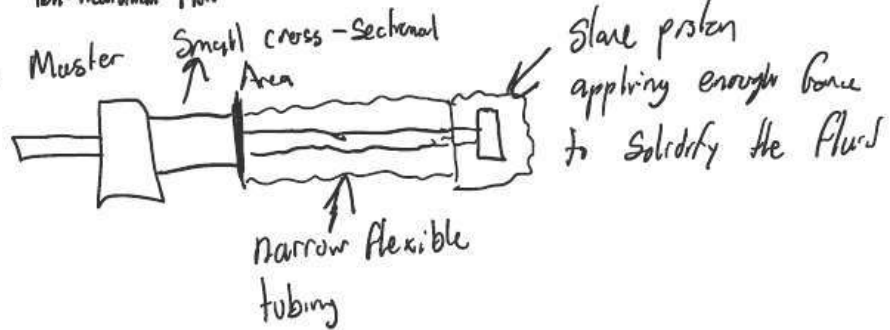
There could be a master piston in the base of the elephant trunk



Narrow area but larger

Multiple straight hard tube so the force can be evenly distributed throughout the fluid

A small controlled force can be applied to the master piston which pushes the slave piston with more force where its enough to solidify the non-newtonian fluid

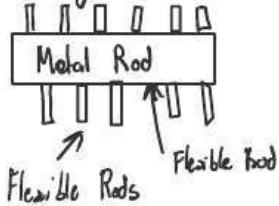


Continuation of Elephant Trunk tip

2. Using Temperature to control the viscosity of the non-newtonian fluid.

Due to fluid mechanics μ (viscosity) $\propto \sqrt{T}$ (Temperature) so increasing/decreasing the temperature,

Design idea will decrease/increase the viscosity.



The metal Rod would be implemented within the silicone tip would be for heating and cooling

In which,

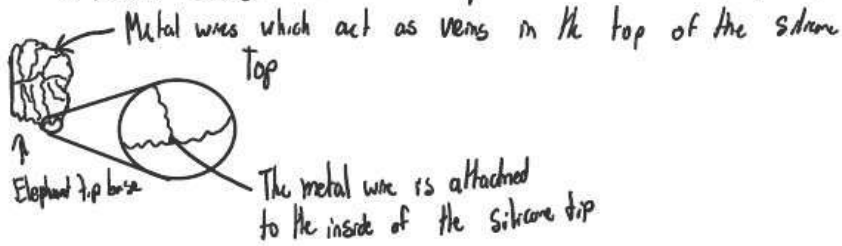
Heating makes the silicone ball liquid which allows to grasp and cup small Objects.

Cooling Solidifies the silicone ball which locks the object

In place.

Continuation of Elephant Trunk tip

3. Another design with the temperature & viscosity principle



∴ The metal wires which are very flexible are inside the non-newtonian substance and electricity is ran through it to increase the heat or make it colder (if it's possible). The reason for the vein like structure is to increase surface area & effectiveness.

The metal wires will have a main flexible but skinny wire and branches where the ends map the inside of the ball to avoid tangling after continuous usage.

Continuation of Elephant Trunk tip

4. Ditching all aspects ~~→~~ focusing on Mechanics (Claw)

We want something very functional and able to access many axes



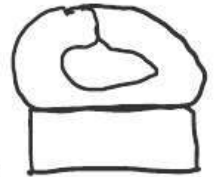
→ we can start off with this as a foundation

Base of the tip

Pros: - grabs stuff

Cons: - limited to one axis

- has a hole in the middle when it closes



Next design



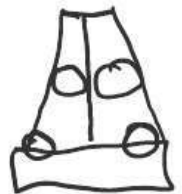
wedge

with this, it eliminates the gap created however it is still limited on one axis.

Pros: - No gap

Cons: - 1 axis

- Extends when trying to push because of the second wedge (joint) in the structure



Next design



wedge

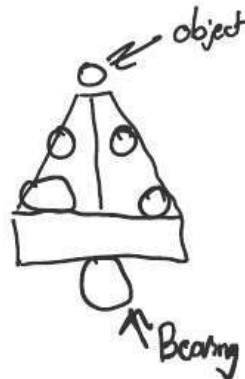
Bearing

With this, the machine has access to all axes 180°

Pros: - more axes

- grabs precise things

Cons: - this only allows for precise grabbing of 1 object only



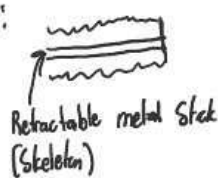
Next design

TBD

Q: How would it pick up a set of marbles all at once???

The trunk itself needs to be designed to act like a snake

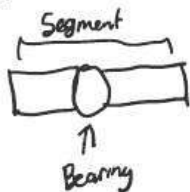
Design:



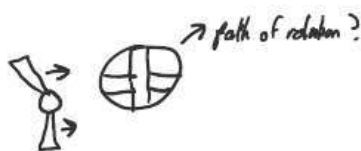
For the main idea we can have a flexible yet retractable metal stick.

Cons: This retraction is purely random if it just pulled
However this is a good start.

If we use the idea of a metal skeleton we would need to program a metal stick to move in 360°



If we have multiple bearings on the rod, all access to axes would be allowed,
we can just program how much each bearing moves



How it works:

open the app and point at an object using the crosshair, The app calculates how much to move and the bearings to gauge how big it is.

Silicone tip material

Things we are looking for, something heat resistant, rubbery, flexible, durable

Heat resistant; this is for if we decide to follow through with the metal wire idea, we'd want to heat up or cool down the non-newtonian material without changing or changing the outside temperature very negligible.

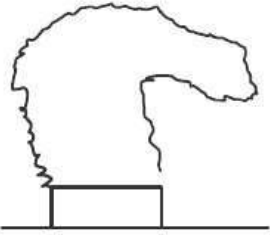
Rubbery; To reduce the risk of damage and increase friction to and with grip. This is purely for the effectiveness as to preserve the life of the metal wires inside as constant temperature change causes wear and tear overtime

Flexible; Pretty self explanatory, to allow for more surface area to pick up small objects, and to allow for more durability as less wear & tear.

Durable; As previously mentioned it needs to be sustainable overtime therefore we need a material that doesn't oxidize or reacts with the metal wiring, and can withstand almost all of nature's elements.

Base design

Ground attachment ; How this works is the device as a whole is put on the ground and mounted. There are many pros & cons with this idea.



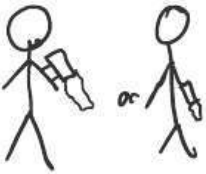
Pros ; - Stable base to hold heavier objects as it uses the ground to anchor

Cons ; - Non portable, stationary

- limited range if in place which basically acts as a sorter then,

- would require more metal to bolt to ground and stabilize

body attachment ; I feel this would be cool to act as a prosthetic for picking up small objects with the idea of the temperature



viscosity silicon top.

Pros ; - Probably the most effective use for a device like this.

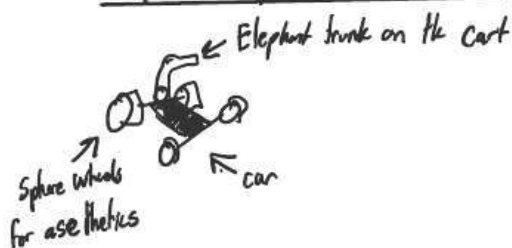
- portable and maybe light

Cons ; - limited to small light objects as the person would only be feeling the weight.

- Needs to be light to allow for comfortability.

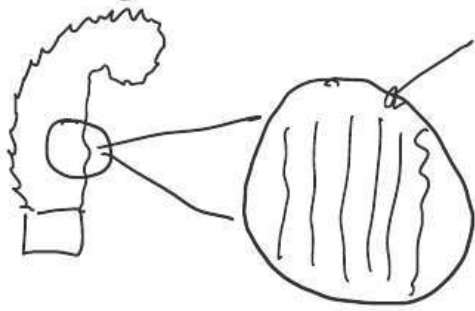
Possible idea:

A portable elephant arm on a roller



← will work on this idea in the next entry.

Redesign: what, if the trunk acted like a real trunk with muscles?



how can we have these muscle fibers more?

Electricity to contract and let go at will

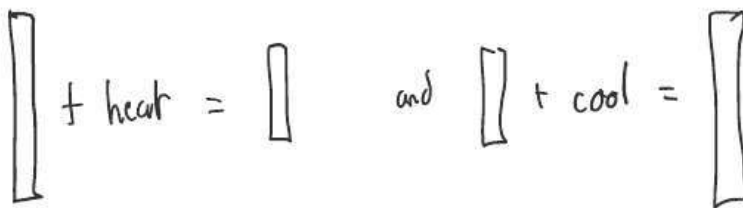
heating equation $P = I^2 R = VI$

wires resistance $R = \rho \frac{L}{A}$

Material?

Nitinol wire is known to contract when heated with electricity time of electricity should be dependent on temp

$$\tau = \frac{P_m c A L}{h A_s}$$



Nitinol only does this via electric currents so

Hypothetically, we specify the currents & where it comes from

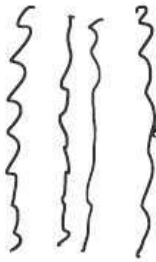
These simulates how the trunk is supposed to act under load

Pros: - more accurate & control

- Can handle more load (tighter grip)

Cons: - Possibly overwear as constant extending and compressing of a metal decreases durability overtime.

What about a change in texture?



outside with metal scales or a trunk that looks like one

What if we add touch sensor skin?

- pressure: detects how hard the trunk touches something

- temperature: detects how hot the object being grabbed is

This could be for ethics or handling something with adequate pressure.

eg. glass or fruit

Controlled with $P = \frac{F}{A}$

or this can make sure nothing living is being held with the arm

and sensitivity $S = \frac{dV_0}{d\rho}$

Material can be regular silicone for grip



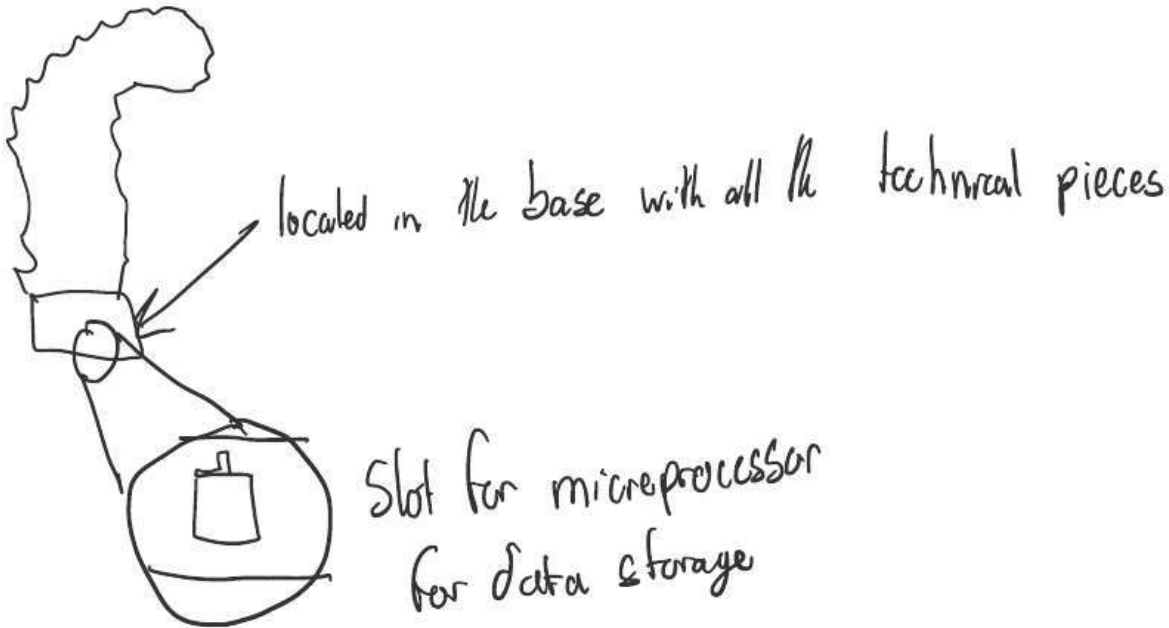
detects object and
gauges temp and how
much pressure to exert

Contact mechanics

on a sphere plane $a = \left(\frac{3FR}{4E^*} \right)^{\frac{1}{3}}$, $P_0 = \frac{3F}{2\pi a^2}$

Let's focus on the trunk's memory and how it adapts

Everytime it picks up something, it must record the force & pressure used and what the shape felt like. Eg. Soft items, slippery, or very fragile



Whats the point?

So it learns more overtime and adapt to improve accuracy

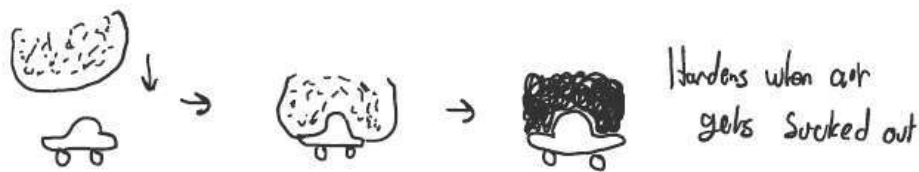
Building on a concept (granular jamming)

When a balloon is filled with grains of sand, it acts as a liquid but the moment air is expelled from the system, it becomes hard as a rock

Using the same principals, we can use this as an easy substitute for

the silicone tip where we have styrofoam beads in a silicone ball.

then as it molts around the object, air gets sucked out & the ball contracts.



Pros: - can handle any small object with care

- glass is easy to pick up as the silicone acts as a gripped hand

We need to keep the durability of the trunk intact for long term. Therefore it should be coated in a coating that heals itself whenever there are small scratches.

layer of the outside material



how it works is if there's a major cut, one of the bubbles in the skin burst and it recoats and seals the cut. almost mimicking actual skin & blood.

microcapsules

Probably injected with epoxy resin?

need a material that heals silicone and reacts with air only. Inside the capsule should not have air

Need to use chemistry to find a substance that reacts with O_2 in the air & manages the reaction time

$$K = A e^{\frac{-E_a}{RT}} \quad \text{or} \quad t = \frac{1}{K}$$

I feel like this should be able to work underwater therefore lets accommodate for it.

for waterproof, we need

- water proof silicone
- Corrosion resistant metal parts
- Very sealed on every gap to prevent water from entering at all

Pros are, this can be for holding cups of liquid and in case there is an accidental spill, the hardware won't be harmed.

Outer skin should have platinum-core silicone which is a military grade underwater silicone
 • strong & durable

→ metal should be titanium as it doesn't corrode

Also must take into account sea pressure (if) this device's main purpose was to be underwater the whole time.

$$P(z) = P_0 + \rho g z$$

At 5m deep

$$P = 101000 + 1000 \cdot 9.8 \cdot 5 \approx 150 \text{ kPa}$$

Also take into account buoyancy

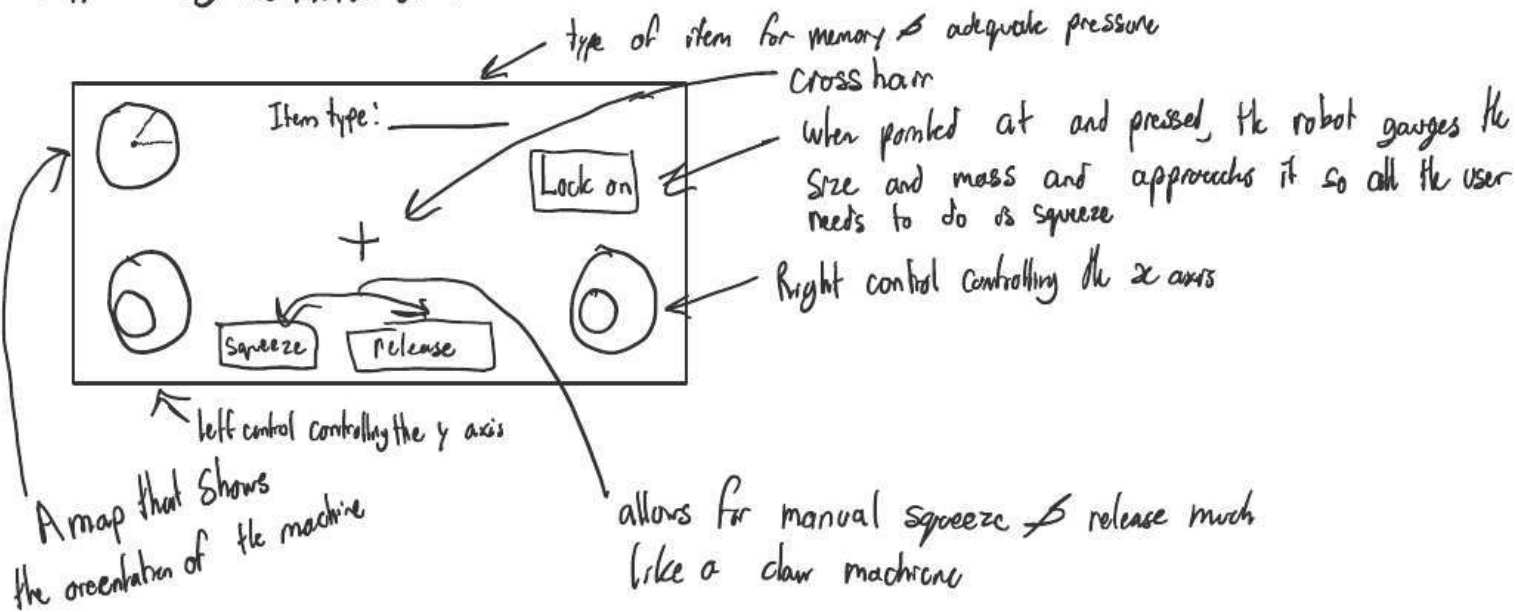
$$F_B = \rho g V$$

$$V = \pi r^2 L$$

Take into account the principle of gradual jamming for the silicone tip

$$F_N \approx \Delta P \cdot A$$

App controlling the motion of the trunk. Needs a good gui:



Basic gui much like any game however its accurate and maybe later can add a "tap to lock on" feature so it automatically orients itself.

Entry 18 - ELEPHANT TRUNK REACHER - Aaron Udho

November 12, 2025 9:11 PM

Settings where the Elephant trunk reacher can be useful

- Firefighting ; - can be heat resistant and used to snake or sift through different debris
 - can have fireman gear attachments so as to spray water or foam
 - can handle explosives from afar or detect survivors
- Agricultural ; - Can use it as a fruit picker to handle soft fruits like berries.
 - can have hoses attached and used as a sprinkler?
- Space maintenance ; - Self explanatory if it was attached on a space craft or space station, outside repair would be done easier.
 - makes it safer for astronauts in the vacuum of space to grab tools in low gravity
- Underwater construction ; - Can help with underwater pipe repairs or coral planting with precision under high water pressure
- Pollution collecting ; - can be attached with a self driving board controlled using AI and go around picking up garbage in populated areas around the world.